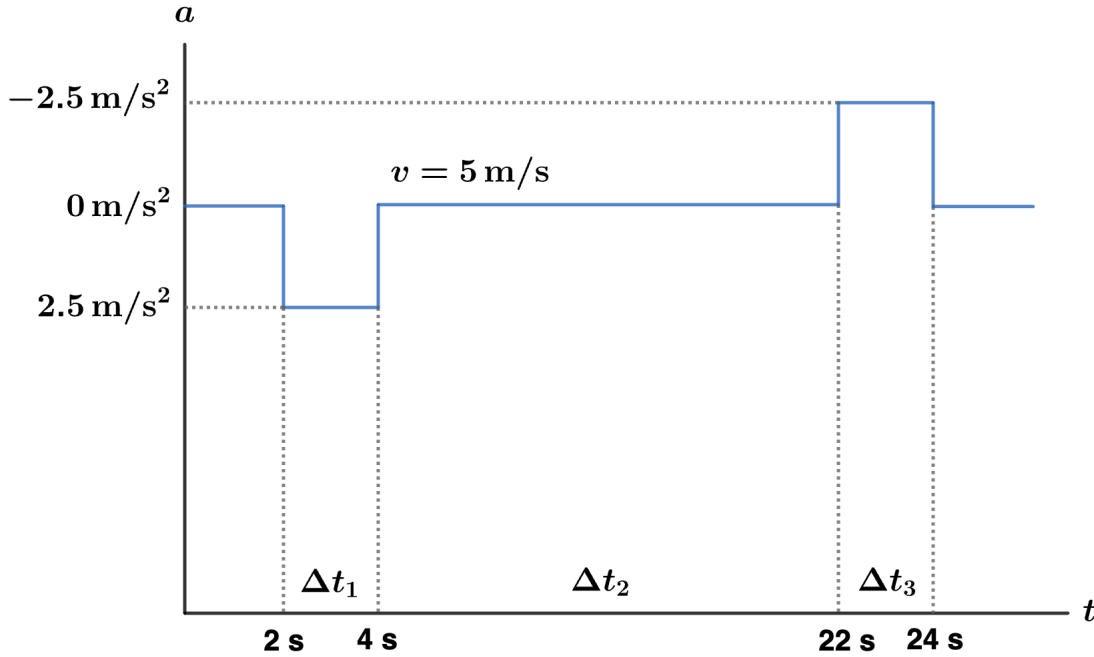


2013 F=ma Exam: Problem 6

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We found in the previous problem the acceleration of the elevator as a function of time:



where we take the positive direction to point downwards. To find the height of the building, we calculate the distance the elevator traveled from beginning to end. Between $t = 2 \text{ s}$ and 4 s ($\Delta t_1 = 4 \text{ s} - 2 \text{ s} = 2 \text{ s}$), the elevator accelerates from rest so

$$d_1 = \frac{1}{2}a(\Delta t_1)^2 = \frac{1}{2}(2.5 \text{ m/s}^2)(2 \text{ s})^2 = 5 \text{ m}$$

At $t = 4 \text{ s}$, the elevator is moving with velocity $v = a\Delta t_1 = (2.5 \text{ m/s}^2)(2 \text{ s}) = 5 \text{ m/s}$. From $t = 4 \text{ s}$ to 22 s ($\Delta t_2 = 22 \text{ s} - 4 \text{ s} = 18 \text{ s}$), the elevator moves at constant velocity so

$$d_2 = v\Delta t_2 = (5 \text{ m/s})(18 \text{ s}) = 90 \text{ m}$$

From $t = 22 \text{ s}$ to $t = 24 \text{ s}$ ($\Delta t_3 = 24 \text{ s} - 22 \text{ s} = 2 \text{ s}$), the elevator decelerates to rest so

$$d_3 = v\Delta t_3 - \frac{1}{2}a(\Delta t_3)^2 = (5 \text{ m/s})(2 \text{ s}) - \frac{1}{2}(2.5 \text{ m/s}^2)(2 \text{ s})^2 = 5 \text{ m}$$

Thus,

$$h = d_1 + d_2 + d_3 = 5 \text{ m} + 90 \text{ m} + 5 \text{ m} = 100 \text{ m}$$

so the answer is C.